

Week 3 · Day 1 · VPC Fundamentals

Create a Custom VPC with Public & Private Subnets

 Time

70 min

 Objective

Build a production-ready custom VPC from scratch with a /16 CIDR block, two public subnets and two private subnets spread across two Availability Zones.

 Prerequisites — Complete Before Starting

IAM admin access · Excalidraw for diagramming

 Portfolio Goal — What You Will Post

Add a VPC architecture diagram to your portfolio GitHub repo. This is the most fundamental networking skill any AWS architect must demonstrate.

Lab Steps

Step 1 Plan Before You Build

1. On paper or Excalidraw, plan your VPC before touching the console. This is a professional habit.
2. VPC CIDR: 10.0.0.0/16 (65,536 IPs total).
3. Public Subnet 1 (AZ-a): 10.0.1.0/24 — web servers, load balancers.
4. Public Subnet 2 (AZ-b): 10.0.2.0/24 — web servers, load balancers.
5. Private Subnet 1 (AZ-a): 10.0.10.0/24 — app servers.
6. Private Subnet 2 (AZ-b): 10.0.20.0/24 — databases.
7. Name your VPC: OluTech-Production-VPC.

Step 2 Create the VPC

1. Go to VPC → Your VPCs → Create VPC.
2. Select VPC only (not VPC and more — we will build manually to learn).
3. Name tag: OluTech-Production-VPC.
4. IPv4 CIDR: 10.0.0.0/16.
5. Tenancy: Default.
6. Click Create VPC.

Step 3 Create the Four Subnets

1. Go to VPC → Subnets → Create subnet.
2. Select your new VPC.
3. Subnet 1: Name: Public-Subnet-AZ-A | AZ: select first AZ | CIDR: 10.0.1.0/24.
4. Click Add new subnet.
5. Subnet 2: Name: Public-Subnet-AZ-B | AZ: select second AZ | CIDR: 10.0.2.0/24.
6. Click Add new subnet.
7. Subnet 3: Name: Private-Subnet-App-AZ-A | AZ: first AZ | CIDR: 10.0.10.0/24.
8. Click Add new subnet.
9. Subnet 4: Name: Private-Subnet-DB-AZ-B | AZ: second AZ | CIDR: 10.0.20.0/24.
10. Click Create subnet.

Step 4 Enable Auto-Assign Public IP on Public Subnets

1. Select Public-Subnet-AZ-A → Actions → Edit subnet settings.
2. Check: Enable auto-assign public IPv4 address. Save.
3. Repeat for Public-Subnet-AZ-B.
4. Leave private subnets as-is (auto-assign OFF).

Step 5 Verify & Diagram

1. Go to VPC → Your VPCs. Confirm OluTech-Production-VPC exists with 10.0.0.0/16.
2. Go to Subnets. Filter by your VPC. Confirm all 4 subnets exist with correct CIDRs.
3. Complete your Excalidraw diagram — add CIDR ranges to each subnet, AZ labels, and VPC boundary box.
4. Export diagram as PNG.



Screenshot Checklist

Take these screenshots and save them to your portfolio folder:

- ☐ VPC console showing OluTech-Production-VPC created
- ☐ Subnets list showing all 4 subnets with correct CIDRs
- ☐ Public subnet settings showing auto-assign public IP enabled
- ☐ Your completed VPC architecture diagram with all labels



Portfolio Post — Copy & Customise

Post your VPC diagram: 'Built my first production-ready custom VPC today — 10.0.0.0/16 CIDR, 4 subnets across 2 AZs, separated into public and private tiers. This is the networking

foundation every AWS application runs on. Diagram attached. #VPC #AWSnetworking #CloudArchitecture'

Bonus Challenge — Push Your Skills Further

Go to VPC → Your VPCs. Find the default VPC that AWS created. Compare its CIDR, subnets, and settings to your custom VPC. Write 5 differences. Why should you never use the default VPC in production?

Lab Complete — Mark Off Your Progress

Lab Steps Completed	☐
All Screenshots Taken	☐
Portfolio Artifact Posted	☐
Bonus Challenge Attempted	☐